

Chapter 2

UV-Ozone Cleaning for Removal of Surface Contaminants

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1 INTRODUCTION

Removal of surface contaminants is essential to all processes where the surface must be modified in some manner, such as deposition of thin films, bonding, or surface patterning. Both organic and inorganic contaminants can cause unreliable bonding or even prevent continuous bonding. Wet and dry cleaning are well-established processes for removal of surface contaminants in a variety of industrial applications. Many of the conventional solvents used for wet cleaning, such as hydrochlorofluorocarbons (HCFCs), are considered detrimental to

the environment and are increasingly subject to regulations for reduction in their use and eventual phase out [1–3]. As a result, there is a continuing effort to find alternate cleaning methods to replace solvent cleaning. Several dry cleaning methods have been developed and have been commercialized using lasers, microabrasives, plasma, ultraviolet (UV)-ozone (UV-O₃), solid gas pellets or soft snow (CO₂, Ar-N₂), electrostatic charge, water ice crystals, micro- and nanoparticle beams, and high-velocity air jets [4–6]. UV-O₃ is a highly effective method for removal of organic surface contaminants to near-atomic levels. Compared to other dry surface treatment techniques, UV-O₃ treatment has the distinctive advantages that it can be conducted under atmospheric pressure, and hence the equipment and running costs are relatively low. Gas plasma (oxygen, hydrogen plasmas), in particular, can have a significant sputtering effect as it contains a complex mixture of protons, electrons, ions, radicals, and excited species with high kinetic energy. In contrast, UV-O₃ cleaning is milder than the oxygen plasma due to the absence of high kinetic energy particles. This means that the UV-O₃ cleaning method can complement conventional cleaning techniques based on oxygen or hydrogen plasma for a variety of applications. UV-O₃ can be used for removal of surface contaminants and for modification of the surface. Several reviews of the technology have been published [7–12]. The intent of this chapter is to provide an overview and discuss recent applications of UV-O₃ cleaning for removal of surface contaminants.

2 SURFACE CONTAMINATION AND CLEANLINESS LEVELS

The most common categories of surface contaminants include particles, thin film or molecular contamination that can be organic or inorganic, ionic contamination, and microbial contamination [13]. Other contaminant categories include metals, toxic and hazardous chemicals, radioactive materials, and biological substances, which are identified for surfaces employed in specific industries. Surface contamination can be in many forms and may be present in a variety of states on the surface. Common contamination sources can include machining oils and greases, hydraulic and cleaning fluids, adhesives, waxes, human contamination, and particulates. In addition, a whole host of other chemical contaminants from a variety of sources may soil a surface. Typical cleaning specifications are based on the amount of specific or characteristic contaminant remaining on the surface after it has been cleaned.

Cleanliness levels in precision technology applications are typically specified for particles by size (in the micrometer (μm) size range) and number of particles, as well as for hydrocarbon contamination represented by nonvolatile residue (NVR) in mass per unit area for surfaces or mass per unit volume for liquids [14–18]. The cleanliness levels are based on contamination levels established in industry standard IEST-STD-CC1246D for particles from Level 1 to Level 1000 and for NVR from Level AA5 (10 ng/0.1 m²) to Level J (25 mg/0.1 m²) [17]. The cleanliness levels have been revised or redesignated in

revision E of this standard [18]. The maximum allowable number of particles for each particle size range has been rounded in revision E, while the NVR designation levels have been replaced with a single letter R followed by the maximum allowable mass of NVR. For example, former NVR level J has the new designation R25; level A/2 is now R5E-1; and level AA5 is now R1E-5. These changes are discussed in greater detail in Volume 7 of this series [13].

Many of the products and manufacturing processes are also sensitive to, or they can even be destroyed by, airborne molecular contaminants (AMCs) that are present due to external, process or otherwise generated sources, making it essential to monitor and control AMCs. AMC is chemical contamination in the form of vapors or aerosols that can be organic or inorganic, and it includes everything from acids and bases to organometallic compounds and dopants [19,20]. A new standard ISO 14644-10, “Cleanrooms and associated controlled environments—Part 10: Classification of surface cleanliness by chemical concentration” [21] is now available as an international standard that defines the classification system for cleanliness of surfaces in cleanrooms with respect to the presence of chemical compounds or elements (including molecules, ions, atoms, and particles).

In many commercial applications, the precision cleanliness level is defined as an organic contaminant level of less than 10 μg of contaminant per cm^2 , although for many applications the requirement is set at 1 $\mu\text{g}/\text{cm}^2$ [17,18]. These cleanliness levels are either very desirable or are required by the function of parts such as medical devices, electronic assemblies, optical and laser components, precision mechanical parts, and computer parts.

3 PRINCIPLES OF UV-OZONE CLEANING

The basic principle of UV- O_3 cleaning for removal of surface contaminants involves the reduction of organic contaminants into harmless H_2O , CO_2 , and NO_x [7–11,22]. At the same time, inorganic contaminants are converted to highly oxidized states that are removed by rinsing with a fluid such as ultrapure water.

UV rays with a short wavelength of 184.9 nm are absorbed by oxygen to form ozone and atomic oxygen. Long-wavelength UV radiation of 253.7 nm is absorbed by ozone to decompose the ozone forming highly reactive atomic oxygen, or it is absorbed by most hydrocarbon substances to decompose them. Thus, the coexistence of wavelengths of 184.9 and 253.7 nm causes continuous formation of atomic oxygen, as well as continuous formation and/or decomposition of ozone. Atomic oxygen formed during the formation and destruction of ozone acts as a strong oxidizer. UV rays with these wavelengths are commonly generated from a commercially available low-pressure mercury vapor lamp, which consists primarily of 253.7 nm radiation, smaller amounts of 184.9 nm radiation, and minor amounts of longer wavelength radiation (Fig. 2.1) [23–25]. However, ozone destruction peaks are around 260 nm, which makes

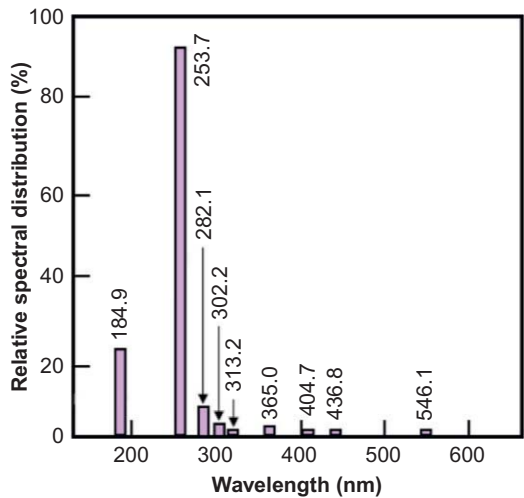


FIGURE 2.1 Spectral distribution of UV radiation in a low-pressure mercury vapor lamp [23–25].

mercury lamps inefficient as sources of ozone production because large portions of generated ozone are simultaneously destroyed.

The energy per mol of UV rays with wavelengths of 184.9 and 253.7 nm is 646.97 and 471.52 kJ/mol, respectively. Table 2.1 shows the bond energies of various molecules of organic compounds [26–29]. Organic compounds can be decomposed by irradiating them with UV rays of these wavelengths if the energy per mol is greater than the bond energy. These excited contaminants, or the free radicals of the contaminants formed by photolysis, react with atomic oxygen to form simple volatile molecules such as CO₂, H₂O, N₂, and O₂, which

TABLE 2.1 Average Bond Energies for Several Molecules [26–29]

Bond	Bond Energy (kJ/mol)	Bond	Bond Energy (kJ/mol)
O—O	143.8	C=C	610.6
O=O	494.4	C≡C	834.6
O—H	463.4	C=O	755.9
C—C	345.7	C—Cl	331.8
C—H	413.5	H—F	565.0
C—N	296.7	C—F	470.4
C≡N	857.7	H—Cl	431.3
C—O	355.7	N—H	363.3

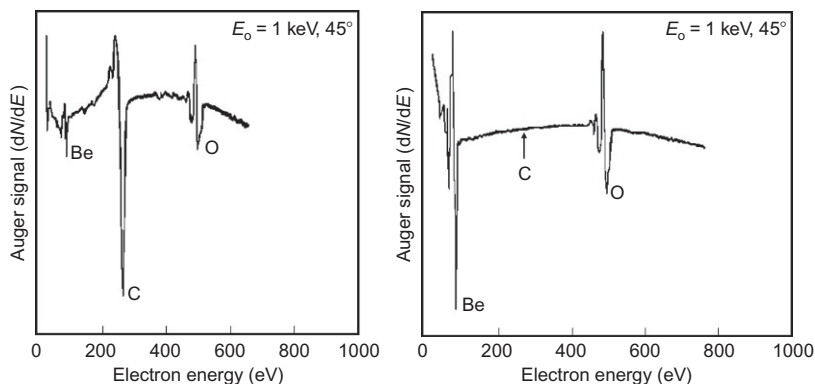


FIGURE 2.2 Auger spectra of a sintered beryllium oxide specimen before (left figure) and after (right figure) cleaning with UV-O₃ for 30 min. The spectra were obtained at 1 keV primary electron energy and at 45° angle of incidence [31].

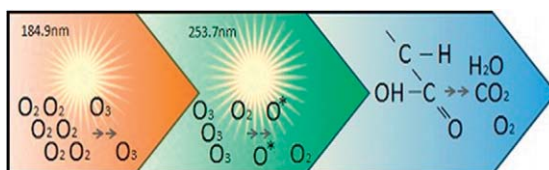
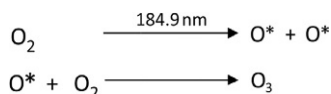


FIGURE 2.3 Principle of UV-ozone cleaning using mercury lamp source [32].

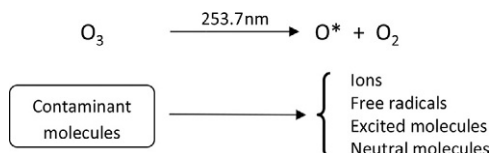
desorb and leave an atomically clean surface. Cleanliness of such UV-O₃ cleaned surfaces has been verified on many occasions by high sensitivity surface analytical techniques, such as electron spectroscopy for chemical analysis, Auger electron spectroscopy, ion scattering spectroscopy/secondary ion mass spectrometry, and total reflection X-ray fluorescence spectrometry (TXRF) [30]. Figure 2.2 shows the Auger spectra of a sintered BeO specimen before and after UV-O₃ cleaning [31].

The three stages of the process are given below and shown schematically in Fig. 2.3.

Stage I Ozone generation: Oxygen in the atmosphere absorbs the UV radiation at wavelength of 184.9 nm generating atomic oxygen. Atomic oxygen reacts with molecular oxygen in the air to form ozone.



Stage II Ozonolysis: The generated ozone absorbs the UV radiation of 253.7 nm and is dissociated to form atomic oxygen. Simultaneously, the hydrocarbon contaminants also absorb UV radiation and are dissociated to form a variety of substances, including excited molecules, free radicals, ions, and neutral molecules.



Stage III Contaminant destruction and removal: The excited substances and the free radicals react with atomic oxygen converting them to simpler volatile molecules H_2O , CO_2 , and NO_x which desorb from the surface.

The processes in the above stages occur simultaneously when both UV wavelengths are present and surface contaminants are removed continuously.

An alternate source of UV radiation is excimer emission that has been developed in the form of xenon lamps for cleaning applications [25,33–52]. Excimers emit in the vacuum UV (VUV) region of the spectrum where the principal absorber is oxygen in the air. The concentration of atomic oxygen and ozone is higher for VUV at 172 nm generated from Xe_2^* dimers. At 172 nm, there is essentially 200% photon efficiency for ozone production in oxygen or air, as nitrogen is transparent to 172 nm radiation. Hence, the cleaning rate and efficiency will be higher at 172 nm than at 184.9/253.7 nm, as shown in Fig. 2.4 for

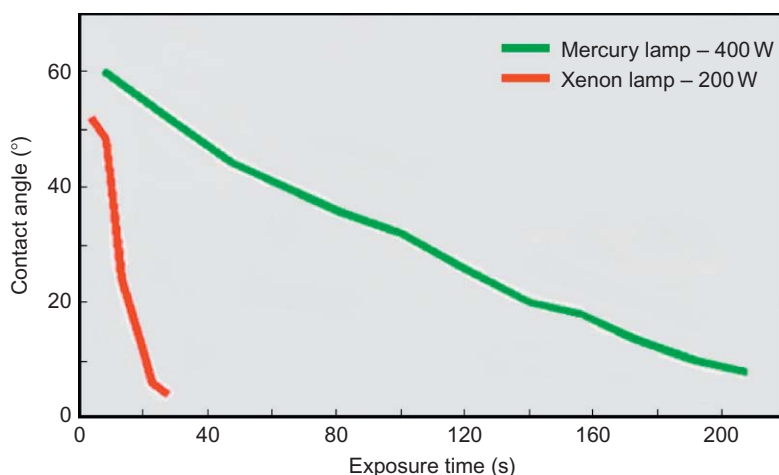


FIGURE 2.4 Comparison of UV cleaning of a silicon wafer with a mercury lamp (184.9/253.7 nm) and a xenon excimer lamp (172 nm) [51].

cleaning a semiconductor wafer [51]. In this case, excimer VUV cleaning is much faster than mercury UV lamp cleaning: the excimer lamp takes 15 s to change the water contact angle from approximately 60° to less than 3° ; the low-pressure mercury lamp takes 220 s at almost twice the power to change the contact angle from 60° to 8° . The energy per mol of 172 nm wavelength UV radiation is 695.49 kJ/mol, which will break all single molecular bonds and several of the double and triple molecular bonds in Table 2.1 with energies less than 695.49 kJ/mol.

The main characteristics of excimer dielectric barrier discharge (DBD) lamps are given below [40,48].

- The optical output spectral distribution of excimer DBD lamps is very narrow. For example, the 172 nm line of the popular Xe_2^* excimer has a full-width at half-maximum value of 12-14 nm. The output is nearly pure UV radiation of high energy at low temperature.
- Since generation of excimers is nearly independent of gas temperature, excimer lamps do not require a warm-up time and may be reignited at any time. In contrast to high-pressure mercury gas discharge lamps, DBD lamps are comparatively cold light sources which in most applications do not need extra cooling. This also minimizes heating and damage to the irradiated surface.
- UV light output is available instantly (in less than 1 ms) which allows for application with repetitive on-off operation of these sources.
- Excimer DBDs can be dimmed by means of pulse density modulation or pulse package density modulation and operated at very low output power.
- The electrodes are not galvanically connected with the gas discharge. Due to low electrode wear, excimer DBD lamps theoretically offer a long lifetime of up to 80,000 h.
- Excimer DBD lamps may be operated with very high (pulse) power densities that potentially exceed those of mercury low-pressure lamps.
- The most efficient excimer DBD lamps operate with noble-gas or noble-gas halogen fillings. They are mercury-free which makes them especially environmentally friendly.
- Depending on the gas or gas mixture used, the VUV/UV output to electrical input efficiency of Xe_2^* excimer DBD lamps can theoretically be as high as 78%. Efficiencies up to 60% have been achieved. This is considerably better than excimer lasers that are typically only 1-3% efficient.
- The spectral output can be tailored by selecting from a variety of excimer gases and gas mixtures or by shifting the radiation spectrum towards longer wavelengths. This makes it possible to target specific chemical species.
- There is minimal damage to the surface. The UV radiation is absorbed in 1- μm thin layer of the surface contaminant.
- Excimer DBD lamps may be constructed in a broad variety of possible geometries. Linear, coaxial, and plane configurations are already available as commercial products or are at the research stage.

- The low-pressure mercury lamp can be used for irradiating three-dimensional (3D) parts because its effective irradiation distance has a wide range of 0-20 mm. By contrast, the 172 nm Xe excimer lamp has a much shorter effective irradiation distance of 0-3 mm, which makes it challenging to process 3D parts.

For cleaning applications, the same process stages discussed above are applicable with excimer UV sources as with mercury lamp sources.

4 PROCESS VARIABLES

There are several key variables of the UV-O₃ process that are critical to effective application of the technique for removal of surface contaminants [7–11,22–25,32,34,41,51–56].

4.1 UV Sources

Two principal UV sources are in commercial use for removal of organic contaminants.

- Low-pressure mercury lamps for 184.9 nm and 253.7 nm wavelength. For most effective cleaning, both 184.9 and 253.7 nm wavelength radiation must be present simultaneously.
- Excimer sources for 172 nm radiation emitted in VUV.

For microbially contaminated surfaces, additional wavelengths from 207 to 370 nm have been employed for removal of the microbial contaminants, using both mercury and excimer lamps [53–56].

4.2 Distance from the Source

Ozone has a broad absorption band around 255 nm with an absorption coefficient of approximately 131 cm/atm [57]. The intensity of the radiation decreases exponentially with the distance of the UV source to the sample. Therefore, the sample should be placed as close to the UV source as possible to maximize the cleaning rate. Typically, the recommended distance is 1-5 mm.

4.3 Precleaning

In general, gross contamination cannot be easily removed with UV-O₃ without precleaning the surface, especially if the contamination contains inorganic salts, dust, or similar substances that are resistant to photosensitized oxidation. These materials cannot be transformed into volatile products by UV-O₃. Precleaning is also recommended for removing thick surface contaminant films that could transform into a UV-resistant film due to the cross-linking action of UV radiation.

4.4 Types of Contaminants

A wide range of contaminants have been successfully cleaned with UV-O₃.

- Cutting oils
- Mixtures of beeswax and pine resin
- Lapping agents
- Vacuum-pump oils
- Silicone diffusion-pump oils
- Silicone vacuum greases
- Soldering fluxes
- Human sebum
- Organic contaminants adsorbed during long-term air exposure
- Carbon thin films formed by vacuum evaporation
- Microbial contaminants (bacteria, fungi)
- Fat on the skin, acid flux, cosmetic greases, resin additives, and waxes
- Solvent residues such as acetone, methanol, and isopropyl alcohol.

4.5 Types of Substrates

The UV-O₃ process has been applied to clean a variety of substrates as pretreatment for subsequent coating and bonding. In general, the adhesion of the surface is improved significantly after UV-O₃ treatment. For example,

- Semiconductor wafers such as silicon, germanium, gallium arsenide, and gallium nitride
- Ceramic materials such as mica, quartz, silicon nitride, and alumina
- Indium oxide and indium tin oxide
- Chromium masks
- Carbon nanotubes
- Glass plates
- Metals including stainless steel, nickel, chromium, platinum-iridium alloys, aluminum alloys, gold, silver, and copper
- Polymers including polyethylene, poly(vinyl chloride), poly(ethylene terephthalate), polystyrene, poly(ether ether ketone), styrene butadiene styrene, and poly(dimethylsiloxane)
- Polyimide composites and aramid films
- Fabrics such as polyester, wool, and silk.

5 CLEANING SYSTEMS

UV-O₃ cleaning systems are commercially available in several sizes and models. [Figure 2.5A–D](#) shows examples of free-standing cleaning units [58–79]. Typically, compact bench top cleaning systems consist of one or more drawer-loaded sample stages that can process parts of various sizes. A normal



(D) (i)



(ii)



Continued

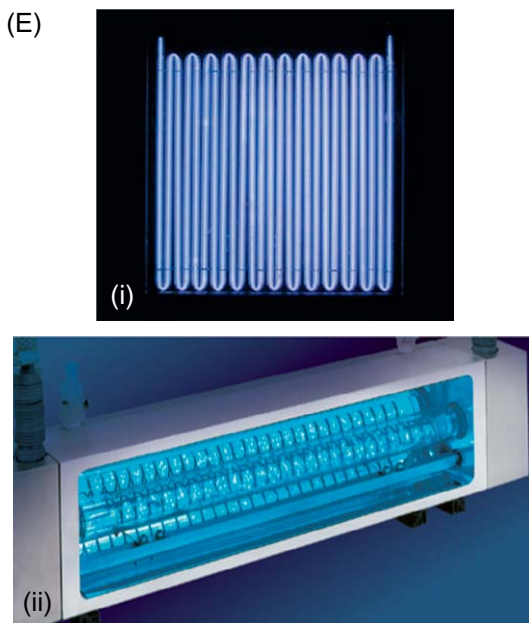


FIGURE 2.5 (A) Single drawer tray UV-O₃ cleaning units. (B) Larger capacity UV-O₃ cleaners. (i) Filgen High Power Cleaner [58], (ii) PSD Pro Series Digital UV-Ozone System [59], and (iii) UVO cleaner [60]. (C) Single and multiple part cleaning units [61–63]. (D) High-end automatic UV-O₃ cleaning units for production applications [61, 77]. (E) Examples of different UV lamps (i) continuous serpentine tube [62] and (ii) discrete tubes [64]. (Panel A: Courtesy of Bioforce Nanosciences. Panel B: (i) Courtesy of Filgen Inc., Japan, (ii) Novascan Technologies, and (iii) Jelight Company, respectively. Panel C: Courtesy of SAMCO, Inc., Japan (i and ii), UVOTECH Systems (iii), and Technovision Inc., Japan (iv). Panel D: (i) Courtesy of SAMCO, Inc., Japan and (ii) Senlights Corporation, Japan. Panel E: (i) Courtesy of Technovision Inc., Japan and (ii) Heraeus Noblelight, Germany..)

UV lamp or a Suprasil[®] lamp is available in different shapes from a continuous serpentine tube (Fig. 2.5Ei) to discrete linear tubes (Fig. 2.5Eii). The Suprasil[®] lamp transmits 90% of 172 nm and higher radiation compared with about 65% transmission with fused quartz source (Fig. 2.6) [64]. This has the advantage of making the cleaning process more effective.

Although ozone can be generated by irradiating oxygen (air) with short wavelength light (184.9 nm; photon energy at this wavelength is 6.70 eV or 154.59 kcal/mol), a separate ozone generator may be required to provide sufficient ozone concentration to clean the contaminants more rapidly, such as ashing of photoresists. The cleaning units are equipped with inlet ports for different gases and an exhaust port to connect to the exhaust system.

High-end semiautomatic and fully automatic systems are available with motorized drawer trays or conveyor sample stages and a microprocessor

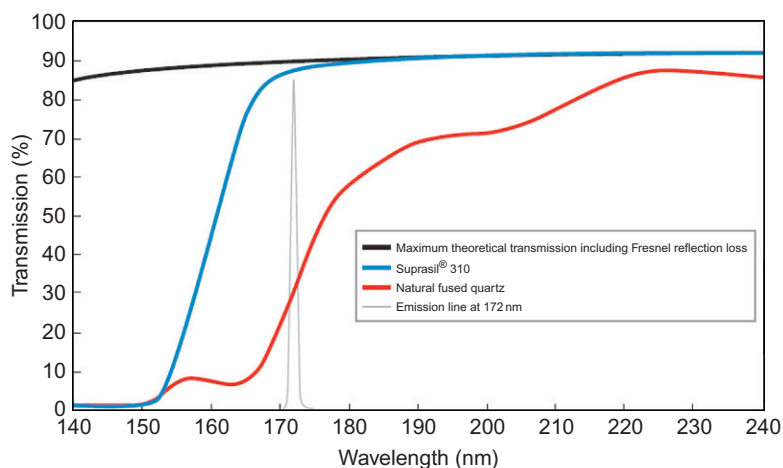


FIGURE 2.6 Typical transmission spectrum of Suprasil[®] glass compared with natural fused quartz [62].

controlled operator interface. The latter systems can be integrated into a controlled environment such as a glovebox or a cleanroom to address the needs in various contamination-controlled manufacturing applications (Fig. 2.7).

UV-O₃ cleaning systems use integrated process as management for gas flow and exhaust. A control unit makes it possible to set the process (irradiation) time, while an integrated flow meter is used to adjust the process gas flow during the cleaning process. Once the preset irradiation time has ended, the oxygen flow is automatically shut off and the process chamber is purged with nitrogen until all process gases have been removed. Integrated safety interlocks are provided to

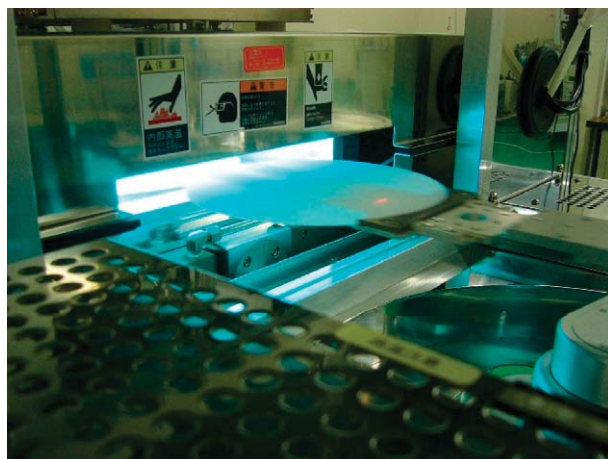


FIGURE 2.7 UV-O₃ cleaner integrated into a cleanroom for automated wafer cleaning [73].

shut off the UV radiation if the door is opened during the process or the exhaust is interrupted.

The process air with ozone is exhausted for safety of operators. However, the concentration of ozone is higher with lower exhaust flow rates (Fig. 2.8), making cleaning and surface modification more effective. An optimum exhaust flow rate is set as a process parameter that balances operator safety and sample surface treatment.

The luminous power of a low-pressure UV lamp is low immediately after lighting, and it could take from 30 s to 5 min until full power is attained. Modern UV-O₃ cleaning systems are equipped with a shutter that separates the lamp from the sample stage, which does not require the lamp to be switched off for operator safety after each sample exposure. A stable amount of UV radiation is always obtained, which enables uniform and highly reproducible process for surface cleaning. Continuous lighting will contribute to extending the lamp life and reduction of the operating cost.

Certain safety and utility features are incorporated in UV-O₃ cleaning systems.

1. Safety interlock switches are attached to the doors. When the doors are opened, the UV lamp turns off automatically.
2. An automatic venting system is coupled to the door interlock switches. The chamber is purged automatically with an inert gas such as nitrogen to eliminate ozone and then ventilated with air to remove the inert gas.
3. The material of construction of the cleaner is selected to be highly resistant to corrosion in the presence of UV and ozone.
4. Organic materials, such as plastic insulation, can degrade when exposed to UV radiation and are not used in the cleaning systems.

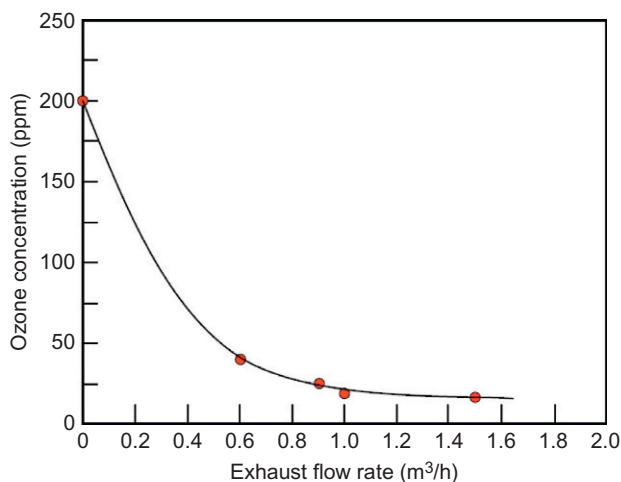


FIGURE 2.8 Ozone concentration increases by decreasing the exhaust air flow rate [75].

5. The cleaning unit is hermetically sealed or completely enclosed to prevent accidental UV exposure and to minimize recontamination by circulating air.

UV-O₃ cleaning systems are very effective and easy to use. The following general guidelines will help maintain optimum cleaning performance of the system.

- The ozone cleaner is not meant for the removal of gross contamination. For the UV-O₃ cleaning procedure to work reliably, the surfaces must be pre-cleaned. One may use a sequence of acetone, methanol, and DI water cleaning with an ultrasonic agitation.
- The best cleaning results are obtained by placing the parts to be cleaned as close as possible to the UV light source. The UV-O₃ cleaner is designed with height adjustable sample stage.
- No acids, bases, chlorinated or fluorinated fumes should be introduced into the UV-O₃ cleaner as they will corrode the internal stainless steel parts and/or damage parts being cleaned.

5.1 Costs

UV-O₃ systems offer several benefits and indirect cost advantages compared with other cleaning techniques. These benefits include improved worker safety and health; reduced environmental liability; no use of toxic chemicals and solvents; eliminated costs of storage, tracking, handling, and disposal of hazardous wastes; and continuous high-throughput production given the short cleaning times. The capital costs are moderate depending on the size and capacity of the cleaning unit (~US\$4000 for a 165 mm wide × 165 mm deep benchtop unit to ~US\$23,000 for a larger bench top unit that can accommodate four times as many parts), but cost of ownership is generally low [58–79]. The primary recurring cost is for replacement of mercury lamp sources which have to be replaced every 3–6 months; excimer lamp sources usually last much longer ~8000 h. Power costs are minimal because the lamps are cold UV sources and, for most applications, no additional cooling is required. However, the cost of mercury recovery (for mercury vapor lamps) and accessory system for destruction of residual ozone have to be considered.

5.1.1 Examples of Cost Savings

Carbon dioxide (CO₂) pellet blasting coupled with UV-O₃ has been investigated in a two-stage process to clean aerospace aluminum alloys on small- and medium-sized components which require a surface cleanliness suitable for bonding [80,81]. Other applications included cleaning of complex parts, the cleaning of nonmetallic parts, and the cleaning of objects prior to the application of optical and thermal protective coatings. For applications in Department of Defense facilities, waste mitigation eliminated more than 155,000 kg of waste annually, which resulted in estimated savings of more than US\$600,000 per

year. Qualitative benefits such as improved mission readiness, improved worker safety and health, reduced environmental liability, uninterrupted production and repair of metal aircraft components, and reduced cost of storage, tracking, handling, and disposal of hazardous waste must also be taken into consideration when assessing the overall benefits of UV-O₃ cleaning.

In a recent study [82], UV-O₃ cleaning for engine pistons was compared with chemical cleaning and with dry ice blasting based on the streamlined life-cycle assessment (SLCA) scores ranging from 0 to 4 (0 has the greatest adverse impact on the environment, 4 has the least impact) [83,84]. Each method is scored for each stage of the process life with respect to materials, energy, and waste (solid, liquid, and gaseous waste). These process stages are listed below.

- Resource provisioning includes all subprocesses involved with preparing the materials and resources needed for the process.
- Process implementation includes all subprocesses involved in setting up the process before operation.
- The primary operation stage includes all subprocesses involved in performing the main function, which is cleaning the piston.
- The secondary operation stage includes all subprocesses that do not perform the main function.
- The end of use stage includes all subprocesses related to disposal, recycling, and refurbishment.

From the assessment, UV-O₃ cleaning achieved the highest score among the three techniques, which makes it the most favorable in its impact to the environment. In further analysis, using the economic input-output life-cycle analysis model [85], the environmental impacts and energy consumption were compared for cleaning 1000 pistons. Again, UV-O₃ cleaning was found to be the most energy and environmentally efficient cleaning method. The cost for cleaning was nearly 20 times less than either chemical cleaning or dry ice blasting. Even when the number of parts is increased to 10,000 or 15,000 pistons, UV-O₃ cleaning qualifies economically as the best cleaning process.

6 ADVANTAGES AND DISADVANTAGES OF UV-OZONE CLEANING

The advantages and disadvantages of UV-O₃ cleaning are given in the following sections.

6.1 Advantages

1. The UV-O₃ cleaning technique is very effective for the removal of molecular level organic contaminants. Ultraclean surfaces can be achieved.

2. This is a dry cleaning process with relatively fast processing times from minutes to hours.
3. There is no damage to the surface. The process can be used on parts with delicate surfaces.
4. Surface cleaning can be performed for effective subsequent processing such as coating and bonding.
5. The cleaning method is very reproducible since UV light intensity and ozone concentration can be accurately measured and reproduced.
6. The system operates at normal temperature and atmospheric pressure. Vacuum pump is not required.
7. The cleaning system is simple and easy to use.
8. The UV light turns on/off instantly. Warm-up operation is not required.
9. UV lamps are air cooled sources. No cooling water is required.
10. No process gas is required.
11. The process does not require fluids with their associated desorption and contamination problems.
12. It has low capital and operating costs. Energy usage is low because of low operating temperature of UV sources.
13. The process is friendly to the environment. No toxic or hazardous chemicals are used. No solvent disposal is required. Mercury can be recovered from used lamps. Excimer lamps do not use mercury.
14. Ozone destruction unit is available as an option.
15. There is little or no down time for maintenance of the system or replacement of parts due to simplicity and high reliability of the equipment.

6.2 Disadvantages

1. Particles and inorganic contaminants cannot be removed as these are not amenable to photosensitive oxidation. The process is limited mainly to removal of biodegradable hydrocarbon contaminants. Most inorganic contaminants, large particles, and other debris cannot be removed.
2. Thick contaminant layers cannot be removed very easily. The removal times are extremely long.
3. Organic materials, such as plastics, are susceptible to degradation in the presence of UV and ozone.
4. Overexposure of some materials such as oxide-forming metals to UV radiation can cause corrosion.
5. Staining and discoloration of materials can result from use of improper wavelengths and exposure times.
6. The process is applicable for final cleaning only. Parts must be precleaned.
7. The low workplace limits for ozone require special design considerations. An ozone evacuation and disposal system is required.

8. Safety precautions are required due to risk of exposure to UV radiation. Even low doses at 254 nm can cause significant damage to the skin.
9. Semiconductor wafer fab tools built to handle large batch sizes using UV-O₃ cleaning process are not readily available.

7 APPLICATIONS

UV-O₃ cleaning has been successfully used for a wide variety of cleaning applications in semiconductor and electronic parts [4–12,86–126], and miscellaneous cleaning applications in optical materials and components, carbon nanotubes, and metals [127–143], polymers [144–156], biomaterials [157–162], masses used as reference standards [163–171], and even recovery of precious metals from ores [172,173].¹ Many patents also have been issued for different aspects of the cleaning process since 1948 when the first patent was issued for removing carbon deposits on engine parts, tools, and painted surfaces [174–189]. It is usually the final cleaning step to achieve contaminant-free surfaces in surface treatment prior to coating, plating, vapor deposition, or bonding. Examples of applications of UV-O₃ cleaning are listed below.

- Atomically clean silicon wafers, GaAs wafers, sensitive lenses, mirrors, and other optical components, solar panels, quartz and ceramic surfaces, cold rolled steel, and inertial guidance subcomponents.
- Cleaning silicon and silicon nitride atomic force microscopy (AFM)/surface probe microscopy probes.
- Cleaning of glasses and glass ceramics components in flat panel liquid crystal display manufacturing, plasma TV production, reticle plate cleaning, quartz oscillator manufacturing, cleaning of optical parts like lenses, prisms and mirrors, piezoelectric buzzers, ceramic large scale integration substrates and other components.
- Metal cleaning during the production of terminal tabs of flat TVs, integrated circuit (IC) wire bonding, micromotor shaft cleaning, laser printer mirror cleaning, semiconductor resin molding die cleaning, and other applications.
- UV cleaning of substrates prior to thin film deposition.
- Maintaining cleaned surfaces during extended storage.
- Cleaning surface plasmon resonance chips and quartz crystal microbalance sensors.
- Cleaning microelectromechanical system (MEMS) and glass devices.
- Cleaning of flux residue from hybrid circuits.
- Ink removal from wafers after testing.
- Cleaning to enhance adhesion of paints, coatings, and adhesives.
- Improved thin film deposition quality.

1. Other common applications of the UV-O₃ process include water filtration, sanitization and disinfection, wastewater treatment, and destruction of highly toxic compounds and other waste forms. These applications will not be addressed in this chapter.

- Stripping photoresist.
- Removing latent images from lithography plates and general cleaning of lithography plates.
- Cleaning circuit boards prior to packaging.
- Cleaning electron and force microscopy samples, surfaces and probes and microscope slides.
- Cleaning and sterilization for biomedical applications.

Some of these cleaning applications are discussed below.

7.1 Semiconductor and Electronics Parts

This is one of the most common application areas for UV-O₃ cleaning, particularly for achieving atomically clean surfaces on wafers. Several thousands of units are in use worldwide and have proven to be cost-effective alternatives to other dry cleaning techniques such as plasma cleaning [12]. Cleaning effectiveness to achieve atomically clean surfaces has been demonstrated (Fig. 2.9) with high-resolution analytical techniques [94,190–196].

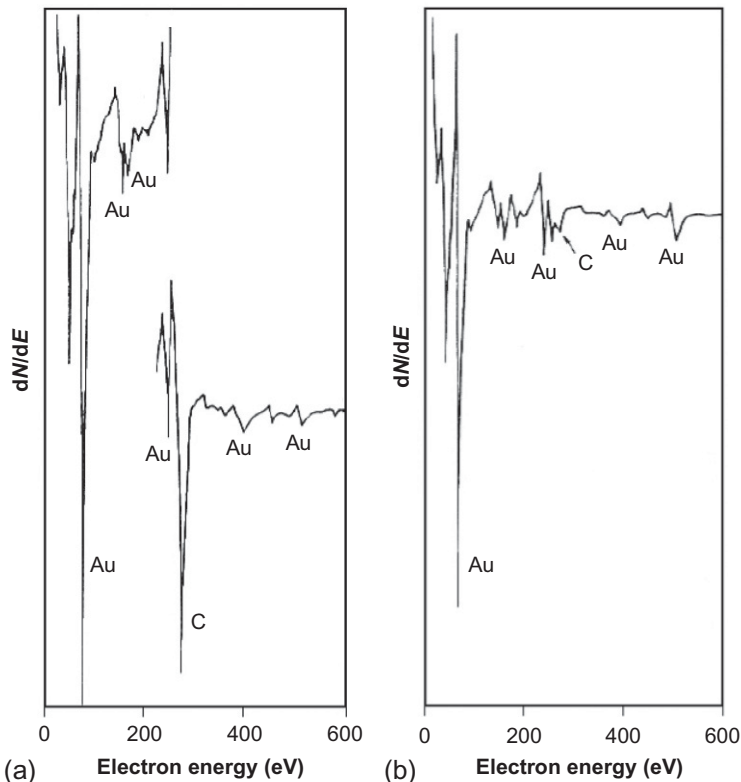


FIGURE 2.9 Auger spectra of a gold surface (a) before and (b) after UV-O₃ cleaning [7].

7.2 Cleaning of Reference Masses

Mass is one of the base SI (abbreviated from Le Système International d'unités) units of measurement. The unit of mass is represented by the kilogram which is defined as the mass of the international Pt-Ir prototype conserved at the Bureau International des Poids et Mesures near Paris. This quantity must be known with the highest accuracy. However, the mass of prototype kilogram standards increases monotonically with time due to atmospheric contamination [167,171]. The main types of contamination are growth of a carbonaceous layer, atmospheric mercury, and sorption of water vapor [163–171]. For storage in air, even a low impingement rate of $20 \text{ ng/m}^2\text{s}$ (the low end of volatile organics typically present in air [197]) represents a mass increase of approximately $500 \text{ }\mu\text{g}$ per hour if all atoms were to adhere to the surface. Similarly, storage in ultrahigh vacuum at $\sim 10^{-9} \text{ Pa}$ hydrocarbon partial pressure, an impingement rate of two three carbon atoms per molecule would lead to higher contamination rate than in air if the sticking probabilities are greater than 0.001 [166]. UV- O_3 cleaning has been successfully demonstrated to remove the carbonaceous layer on exposure of a Pt-Ir mass standard for 75 min, based on predicted $80 \text{ }\mu\text{g}$ mass gain over 100 years exposure to filtered air. The mass of the standard was $372 \pm 6.2 \text{ }\mu\text{g}$ relative to 1 kg before cleaning, which decreased to $295 \pm 6.2 \text{ }\mu\text{g}$ relative to 1 kg after cleaning, showing removal of $77 \text{ }\mu\text{g}$ of contamination.

7.3 Solar Wind Sample Collectors

The NASA Genesis mission was launched in 2001 to collect and return solar wind samples to Earth for analysis of isotopic and elemental compositions [198]. The samples were collected by passively exposing ultrapure materials to the solar wind, including diamond, silicon carbide, diamond-like carbon films, aluminum, Al6061, aluminum alloy, molybdenum, silicon, germanium, gold, sapphire, carbon-cobalt-gold composite film on sapphire, and Zr-Nb-Cu-Ni-Al bulk metallic glass ($\text{Zr}_{58.5}\text{Nb}_{2.8}\text{Cu}_{15.6}\text{Ni}_{12.8}\text{Al}_{10.3}$ composition in atomic percent) [199]. UV- O_3 cleaning was the best nondestructive method for removing carbon contamination from the surface of the Genesis materials before and after the mission to levels below nonflight standards [200–202]. For example, after 30 min treatment by UV- O_3 carbon on the surface was significantly reduced by 70–85% as shown by XPS and TXRF analysis.

7.4 Metal Surfaces

Not only is UV- O_3 cleaning effective on metal parts, it will not damage non-metal components that may be attached to the parts being cleaned such as rubber or plastic fittings. Metal, plastic, and fiberglass parts will maintain their original finish, depending on exposure time. Some surfaces will be cleaned at different rates than others due to the degree and type of contamination on the surface, as

well accessibility of the surface to UV radiation and/or ozone. For example, the inside surfaces of long aluminum tubes (20 cm long, 1 cm diameter) could be cleaned effectively after 20–30 min exposure in a commercial unmodified by UV-O₃ system [127]. No UV radiation could reach the inside surfaces except those at the innermost lip of the tube. However, even at the farthest distance of 20 cm from the tube opening, significant levels (>41%) of carbon contamination could be removed by the process, presumably as the result of direct reaction of O₃ with the surface.

Corrosion of aluminum bonding pads can occur due to the presence of an electrolyte around the pad. Exposure to UV-O₃ improves the adhesion between the molding compound and the integrated circuit chip, which prevents moisture concentration and electrolyte formation at the pad. UV-O₃ cleaning has been shown to be effective in preventing corrosion [134]. The water contact angle was measured immediately after cleaning, and after 18 h and after 24 h in storage of the cleaned parts. Longer cleaning time resulted in smaller contact angle. When it was measured immediately after UV/ozone cleaning, 5 min cleaning reduced the contact angle from 68° to 30°, while cleaning for 10 min reduced the contact angle from 68° to 10°. However, the contact angle measured on cleaned parts in storage decreased from 68° to 25° even after 10 min cleaning, indicating readsorption of contaminants from the air in storage.

Sulfur readily adheres to gold and freshly deposited gold samples exposed to ambient conditions are easily contaminated with any sulfur compounds present in the environment. Bonding of sulfur to gold via self-assembled monolayers of alkanethiols is of considerable interest for contact printing and lithography applications of organic thin films and characterizing the sulfur from such thiol films is critical to understanding their properties and interactions [203,204]. Alkylthiolates bound to gold are converted to alkylsulfonates on irradiation with UV light, which can be removed by rinsing with water [136,205–207]. This principle has been employed as the basis for UV-O₃ cleaning to eliminate sulfur impurities from gold exposed to ambient conditions, as well as for removing previous thiol monolayers to generate a fresh gold surface [132,133]. Five minutes of UV-O₃ exposure was found to be sufficient for routine cleaning of gold.

7.5 Glass and Optical Materials

A glass surface with less than a single molecular layer of an organic contaminant is considered ultraclean. Even in cleanrooms, there are volatile organic compounds and sulfur compounds present in the air. If the cleaned glass is left unprotected in this environment it will get recontaminated. The contact angle of an ultraclean surface will return to 20° after 30–60 min exposure to the environment. An ultraclean surface cannot be maintained ultraclean for extended times and periodic recleaning is necessary. Figure 2.10 shows a glass plate before and after exposure to UV-O₃ for 1 min. Contamination present on the surface

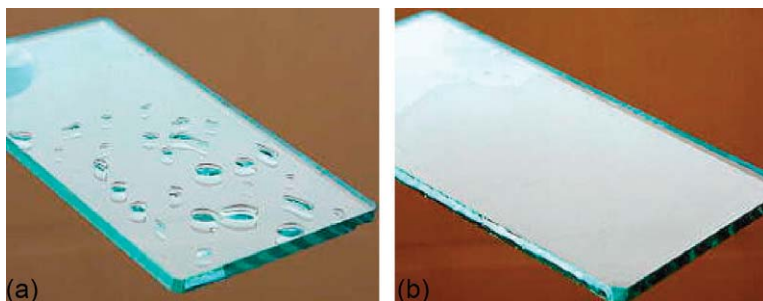


FIGURE 2.10 The surface of a glass plate before (a) and after (b) UV-O₃ cleaning for 1 min [75]. Water beads upon the noncleaned surface, but not on the cleaned surface.

(indicated by water beading on the surface in Fig. 2.10a) was effectively removed (Fig. 2.10b). By increasing the cleanliness of the surface, a well-adhered coating can be achieved without adding expensive extra additives that may reduce the overall quality of the coating.

A new mode of operation using a Xe₂⁺ DBD source has been developed which gives short-pulsed (SP), high peak-power, spatially uniform VUV radiation output [37,41,140]. The SP DBD source was tested in several cleaning applications, including removing optical mount contaminants from surfaces of optical parts, for removing hydrocarbon contamination from optical and polymer surfaces, and dehydroxylation of fused silica. The results showed near-complete removal of hydrocarbon contaminants from the surface after 1-3 h exposure. Contact angles of less than 10° were measured.

7.6 Probe Tips

UV-O₃ cleaning is an easy to use, effective, and *in situ* cleaning technique for cleaning AFM probes prior to starting an AFM experiment, or even during an experiment when the probe tip is contaminated as evidenced by unstable and degraded images. Five to ten minutes of UV-O₃ cleaning are sufficient to remove the hydrocarbon contaminants present on the surface [137,138,142,148,208].

7.7 Decontamination of Incubator Cabinets

Microbial contamination of cultures employed in biological work represents a disruptive factor that requires repetition of complex operations and may also result in the loss of expensive, nonreproducible parent cultures. The temperature and the high level of humidity are ideal conditions for growth not only for the cultures but also for microbial contaminants such as bacteria and fungi. Thus, incubation in CO₂ incubators involves a high risk of contamination; these incubators maintain a constant CO₂ level. No matter how carefully the work is carried out under sterile conditions, contamination of these incubators is

unavoidable, since these airborne microbial contaminants enter the incubator whenever it is opened. The risk of the cultures becoming contaminated is directly dependent on the amount of germ contamination in the incubator. UV-O₃ has been used to successfully decontaminate incubator cabinets using a commercial decontamination unit [157,158]. The unit was placed directly in the incubator cabinet and decontamination treatment was carried out for 2 h per day for a period of 45 days. The average germ level in the UV-O₃ treated incubator was only 20% of the germ level in an untreated incubator.

7.8 Preparation of Samples for Trace Element Analysis

Trace element analysis in biological samples provides information on the presence and chemical state of each element which is important for novel cancer therapy and other medical applications, as well as biomineralization studies [209–211]. Ashing is a procedure often used to remove organic carbon (and other volatile species). This has the effects of thinning the sample (tissue sections thicker than 5 μm can be imaged without charging, after ashing) and of greatly enhancing the relative concentrations of all remaining elements, making it easier to analyze the elements. The elements to be analyzed must be present at a concentration above the detection limit of the analytical technique. UV-O₃ has been successfully demonstrated as an ashing technique for preparation of cell and tissue samples for spectromicroscopic analysis [212–215]. No detectable displacement of material or element redistribution takes place during UV-O₃ ashing at room temperature and atmospheric pressure. Typical ashing times of tissue sections range between 6 h for 50 nm sections to 4 days for 10 μm sections at a distance of less than 5 mm from the UV radiation source. Elements that have been successfully analyzed at trace levels include calcium, potassium, sulfur, phosphorous, gadolinium, copper, and manganese whose relative concentrations were enhanced by UV-O₃ ashing. In fact, calcium, sulfur, and potassium were detectable only after ashing.

7.9 Polymer Surfaces

Many polymer materials display a combination of superior mechanical properties and chemical stability, and are easy to process. However, these same polymers often do not possess the surface properties necessary for their successful application in various fields such as adhesion, coatings, thin-film technology, microelectronic devices, nanotechnology, biomaterials and biotechnology, friction and wear, membrane filtration, and composites. UV-O₃ offers the dual advantage of surface modification of the polymers and cleaning the surface of unwanted contaminants from processing of the polymers [144–156]. Depending on the polymer, exposure times ranged from 1 min to 60 min to achieve a clean surface as well as the desired surface properties, such as improved wettability and adhesion, hydrophilicity, and increased surface energy.

7.10 Textiles and Fabrics

Dyeing of synthetic and natural fabrics, such as wool, silk, and polyester, is a critical finishing step in the processing of these materials. By suitable surface modification, the dyeability of the fabrics can be greatly improved. UV-O₃ radiation has been very successfully used for surface modification of thermally sensitive fabrics because the UV-O₃ treatment can be carried out continuously under atmospheric pressure using simple and inexpensive equipment [149–152,154,156]. Treatment times ranging from 1 to 20 min resulted in significantly increased dyeability, based on color strength, reflectance, and color fastness of the dyes. This increase is attributed primarily to the enhanced hydrophilicity of the surface, and the strong electrostatic attraction between the dyes and the surface of the fabric.

8 SUMMARY

UV-ozone is an effective method for removal of thin film-type surface contaminants from a variety of materials. However, thick contaminant layers cannot be removed and a precleaning step is necessary. UV-O₃ is often the final step to achieve near-atomically clean surfaces for subsequent processing activities such as deposition and bonding. The method is based on UV excitation of surface species and conversion of molecular oxygen to ozone and atomic oxygen. These strong oxidizers then decompose UV excited organic surface contaminants to harmless volatile compounds such as H₂O, CO, CO₂, and N₂ that desorb from the surface, and inorganic contaminants which are converted to highly oxidized states that can be readily removed by rinsing with a fluid such as ultrapure water. The process can be conducted at normal temperature and atmospheric pressure with low capital and operating costs. It is environmentally friendly without the use of toxic solvents or hazardous chemicals requiring expensive management and disposal. Typical applications include cleaning semiconductor and electronic parts, optical materials and components, and carbon nanotubes, metal cleaning, cleaning probe tips, cleaning of polymers and biomaterials, cleaning of masses used as reference standards, decontamination of microbially contaminated incubator cabinets, and ashing of biological samples for trace element analysis.

ACKNOWLEDGMENT

The author would like to thank the members of the STI Library at the Johnson Space Center for help with locating obscure reference articles.

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